Journal of Clinical Epidemiology 139 (2021) 49–56

Journal of Clinical Epidemiology

ORIGINAL ARTICLE

GRADE guidelines 33: Addressing imprecision in a network meta-analysis

Romina Brignardello-Petersen[a,*], Gordon H. Guyatt[a], Reem A. Mustafa[a,b], Derek K. Chu[a], Monica Hultcrantz[c], Holger J. Schünemann[a,d], George Tomlinson[e]

[a] *Department of Health Research Methods, Evidence and Impact, McMaster University, Hamilton, Ontario, L8S 4L8, Canada*

[b] *Department of Internal Medicine, Division of Nephrology and Hypertension, University of Kansas Medical Center, Kansas City, KS 66160, United States*

[c] *Swedish Agency on Health Technology Assessment and Assessment of Social Services (SBU), Stockholm, Sweden*

[d] *Department of Medicine & Institut für Evidence in Medicine, Medical Center & Faculty of Medicine, University of Freiburg, Freiburg, 79110, Germany*

[e] *Department of Medicine, University Health Network, Toronto, Ontario, M5G 2C4, Canada*

Received 11 December 2020; Received in revised form 11 June 2021; Accepted 15 July 2021; Available online 19 July 2021

Abstract

Objective: This article describes GRADE guidance for assessing imprecision when rating the certainty of the evidence from network meta-analysis.

Study Design and Setting: A project group within the GRADE working group conducted iterative discussions, computer simulations, and presentations at GRADE working group meetings to produce and obtain approval for this guidance.

Results: When addressing imprecision of a network estimate, reviewers should consider the 95% confidence or credible interval, and the optimal information size. If the 95% confidence or credible interval crosses a pre-specified threshold, reviewers should rate down the certainty of the evidence. If the 95% confidence interval does not cross any pre-specfied threshold, reviewers should consider the optimal information size. Because addressing the optimal information size may be challenging, reviewers can use the effect size to decide if any calculations are necessary. When the size of the effect is modest or the optimal information size is met, reviewers should not rate down for imprecision.

Conclusion: Reviewers should use this guidance when to appropriately address imprecision in the context of the assessment of certainty of evidence from network meta-analysis. © 2021 Elsevier Inc. All rights reserved.

Keywords: GRADE guidance; Certainty of evidence; Network meta-analysis; Imprecision; Optimal information size

Conflict of interest statement: None.

* Corresponding author. +1 905 5259140 ext 21635
E-mail address: brignarr@mcmaster.ca (R. Brignardello-Petersen).

https://doi.org/10.1016/j.jclinepi.2021.07.011
0895-4356/© 2021 Elsevier Inc. All rights reserved.

What is new?

Key findings

- When addressing imprecision of network estimates, reviewers should consider the 95% confidence interval and its relation with thresholds of interest and the optimal information size.
- The imprecision of network estimates can be serious enough to rate down the certainty of the evidence 3 levels.

What this adds to what was known?

- We describe a sequential process for reviewers to follow when making this assessment, to maximize efficiency and ensure that no necessary step is ignored.
- We provide a previously unavailable approach to address optimal information size in the context of network meta-analysis.

What is the implication and what should change now?

- When assessing the certainty of network estimates, systematic reviewers and guideline developers addressing imprecision should use this approach and be more transparent in the justification of their judgments.

1. Introduction

The Grading of Recommendations Assessments, Development, and Evaluation (GRADE) working group has previously presented guidance for evaluating the certainty of the evidence (confidence in estimates of effect, quality of evidence) in network meta-analysis (NMA) [1-4] and how to interpret its results and draw conclusions [5,6]. This article provides guidance for addressing imprecision of NMA estimates. The discussion assumes familiarity with the basic concepts of NMA and of addressing imprecision in the context of traditional pairwise meta-analysis, and constitutes official guidance from the GRADE working group.

Several factors play a key role in the assessment of imprecision both in traditional pairwise meta-analysis [7] and NMA. These include the degree of contextualization of the rating of certainty of the evidence and the closely related issue of the target of the certainty rating (i.e., what it is in which are we rating our certainty); decisions regarding the thresholds of interest necessary for this assessment; and value judgments regarding the magnitude of effect estimates (e.g., what are a small but important effect, a modest effect, and a large effect). The scope of this article is to illustrate how to assess imprecision after making these decisions.

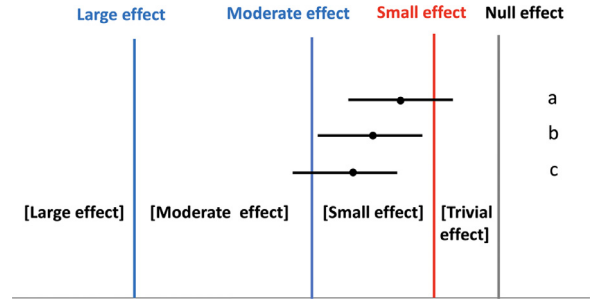

Fig. 1. The relationship between the confidence interval (CI) of an effect and the thresholds determine if reviewers should rate down for imprecision. Vertical lines represent thresholds, which delimit ranges of magnitudes of effect labeled as per in the text between the square brackets. Using a minimally contextualized approach, reviewers would not rate down due to imprecision in A, B, or C if they are rating the certainty that there is any benefit (the CI does not cross the threshold of null effect). They would, however, rate down in A but not in B or C if they are rating the certainty that there is at least a small benefit (the CI in A crosses the small effect threshold, but the CI in B or C does not). Using a partially contextualized approach, reviewers would rate down the certainty of the evidence due to imprecision in A or C but not in B if they are rating their certainty that the effect is small (the CI of A crosses the small effect threshold and the CI of C crosses the moderate effect threshold, but the CI of B does not cross either the small effect or the moderate effect threshold).

2. Addressing imprecision in traditional pairwise meta-analysis

When addressing imprecision in traditional pairwise meta-analysis, systematic reviewers and guideline developers consider the bounds of the confidence interval (CI) of the absolute estimate of effect, and the optimal information size (OIS) [7]. The certainty of the evidence is decreased when the CI crosses specific thresholds of interest or when the OIS is not met. We explain both concepts below and for simplicity, we refer to thresholds of interest as thresholds.

If the CI lies completely on one side of a threshold or between two thresholds, we can make more confident statements regarding the patient-importance of the effect [8]. The thresholds depend on the purpose of the systematic review, and thus on the degree of contextualization [9]. Fig. 1 illustrates the fundamental logic of these decisions.

The CIs estimated based on small numbers of trial participants or few events can be fragile, in the sense that small amounts of additional data can substantially change the location of the interval. The OIS helps address this concern. The OIS is the number of participants obtained with a conventional sample size calculation for a trial addressing the question of interest. When the number of patients included in a meta-analysis for an outcome is larger than the OIS, important changes in the results in the face of considerable amounts of new evidence are unlikely. Initial GRADE guidance advises systematic reviewers to always consider the OIS and for guideline developers to consider OIS when the CI does not cross a threshold[7]. Further

developments of this guidance [9] provide the conceptual underpinning for this paper.

3. The guidance for pairwise meta-analysis may not always be helpful in the context of NMA

When considering the OIS in the context of pairwise meta-analysis, systematic reviewers calculate the total number of participants included across studies. In the context of NMA, because network estimates are calculated using direct and indirect evidence, this approach for determining whether the total sample size meets the OIS often does not work. For a specific comparison, if the CI of the network estimate is similar to the CI of the direct estimate, then there is little information gained from the indirect estimate, and reviewers could add up the number of participants in the trials that compare the two interventions directly to determine if the sample size is sufficient to meet the OIS. When, however, the indirect evidence makes a non-negligible contribution to the network estimate, and the direct and indirect evidence yield similar results (i.e., are coherent), considering only the direct evidence would underestimate the number of participants contributing to the network estimate. In the presence of incoherence, the network estimate will have a wider CI than the direct estimate, and counting the number of people contributing the direct evidence will overestimate the effective sample size of the network estimate.

Fortunately, in many situations considering the OIS may not be necessary. First, if the CI of the network estimate crosses a threshold, reviewers or guideline developers should rate down the certainty of the evidence for imprecision; considering the OIS will not change this decision. Second, if the point estimate of the relative effect indicates a modest effect and the CI does not cross a threshold, one can infer that the sample size is large enough that new evidence is unlikely to result in important changes in inferences. In the subsequent discussion, we elaborate on both these situations.

Considering the above, specific guidance for addressing imprecision in the context of NMA is needed. This guidance will facilitate assessments, help authors of NMAs to achieve explicitness regarding their methods, and users to understand ratings of certainty of evidence.

4. Addressing imprecision in NMA

Fig. 2 summarizes GRADE guidance on how to address imprecision for each network estimate.

4.1. If the CI crosses a threshold, we should rate down for imprecision

Systematic reviewers should choose thresholds based on the degree of contextualization of their review [9]. Using a minimally contextualized approach, reviewers use the null effect to rate their certainty that there is a benefit or a harm. Alternatively, they may choose a small but important effect as their threshold. Using a partially contextualized approach, reviewers use thresholds for small, moderate, and large effects and rate their certainty that the true effect has a particular magnitude (e.g., trivial or none, small, moderate, or large) [10]. Guideline developers using a fully contextualized approach use a threshold that represents an effect large enough to mandate a decision between alternatives. Whatever the degree of contextualization, when the CI crosses one or more of the thresholds, reviewers should rate down the certainty of the evidence (i.e., whenever a CI crosses a threshold, we are less certain that the true effect lies above or below the threshold, or within two thresholds) [9]. This principle applies to both pairwise meta-analysis and NMA estimates (Fig. 1).

Ideally, reviewers would decide a priori where to set the thresholds based on absolute effects. In addition, these thresholds should be consistent with the approach that reviewers will use when interpreting the results of the NMA [5,6]. Because the choice of thresholds must be done using absolute estimates of effect, assessing whether the CI crosses a threshold requires using the CI of the absolute estimate of effect.

For example, in an NMA of methotrexate monotherapy vs. methotrexate combinations for patients with rheumatoid arthritis, the absolute difference between the number of patients who achieved a response when comparing methotrexate plus ciclosporin vs. methotrexate alone was 134 more per 1,000 patients, with a 95% CI from 35 fewer to 290 more [11]. When rating the certainty that there is a benefit (i.e., higher proportion of response with the combination therapy), because the CI crosses the null effect, the authors rated down the certainty of evidence due to imprecision (Fig. 2, scenario 1a).

It is important to highlight that whether a threshold is crossed and the certainty of the evidence should be rated down for imprecision depends on the degree of contextualization, the target of the certainty rating, and thresholds chosen [10]. For example, crossing the null effect threshold is not a reason to rate down the certainty of evidence when the target of the certainty rating is to learn if there is trivial to no effect, and the 95% CI is entirely within the thresholds of small benefit and small harm. Similarly, when rating the certainty of the same estimate, systematic review authors who choose to not make value judgments, those who choose a minimally important difference, and guideline developers who choose a decision threshold may end up with different judgments with regards to imprecision.

4.2. When the CI crosses a threshold, we can rate down the certainty of the evidence one, two, or three levels

Whether reviewers' rate down the certainty of the evidence one or two levels due to imprecision remains a

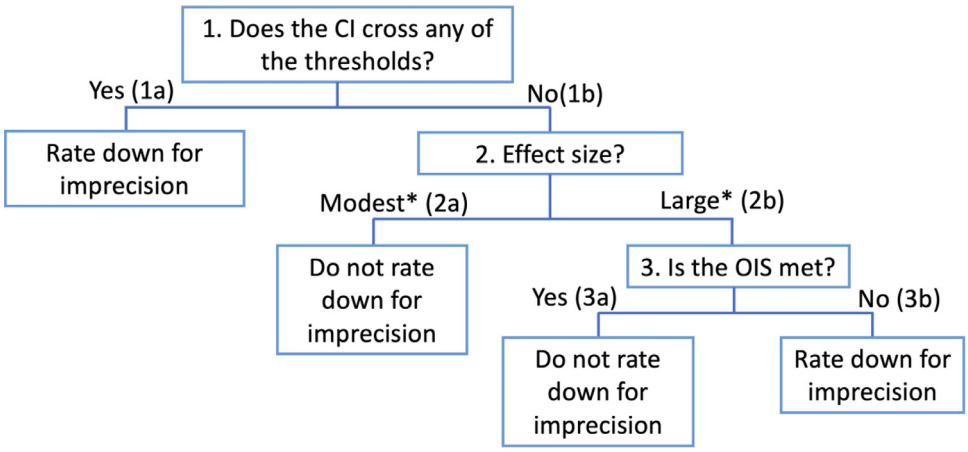

Fig. 2. Process for addressing imprecision for each network estimate. Although the effect size (2) is another way of addressing whether the OIS is met, (3) we illustrate it as a different step because it may help reviewers avoiding to go through the calculations required when addressing 2b. What is a "modest" effect (2) requires value judgment specific to each scenario (an example is 30% relative risk reduction). Step 3 can be addressed by calculating the sample size underlying a network estimate and contrasting this with the optimal information size, or, for OR, and RR effect measures by looking at the ratio of the upper to the lower extremes of the confidence interval.

matter of judgment. It depends on where the extremes of the CI are and how many thresholds it crosses, as well as the interpretation of these extremes in absolute terms. For continuous outcomes, it also depends on the outcome, and the scale of measurement.

It may be helpful, however, to consider the consequences of rating down one vs. two levels. For example, in an NMA of pharmacological interventions for acute diarrhea and gastroenteritis in children, when comparing Kaolin Pectin vs. standard care, the reduction in the number of hours of diarrhea duration was 5 (95% CI, reduction of 34 hours to increase of 23 hours) [12]. Assuming that imprecision was the only serious concern, reviewers following GRADE guidance on how to communicate results [13] would, if they rated down one level, have concluded that there is "probably" or "likely" a reduction in the duration of diarrhea when using Kaolin Pectin. If they rated down two levels, however, they would have concluded that there "may" be a reduction in this outcome. The latter seems more appropriate, thus illustrating how bearing in mind the consequences of rating down one vs. two levels can facilitate the judgment.

In addition, there are situations in which rating down three levels due to imprecision may be appropriate. In one of the first iterations of a living NMA of pharmacological treatments for Coronavirus Disease 2019 [14], several network estimates had extremely imprecise CIs. For example, the risk difference in mortality per 1,000 patients was 329 fewer (95% 330 fewer to 670 more) when comparing ribavirin to standard care. This CI provides very little information about the effect of ribavirin on mortality because it crosses multiple thresholds and its bounds reflect dramatically different effects (the lower bound of 330 fewer deaths represents a large benefit, and the upper bound of 670 more deaths represents a large harm). Even assuming no other serious concerns in this body of evidence, concluding that there may be a benefit would be misleading. Thus, reviewers could rate down the certainty of the evidence 3 levels resulting in a rating of very low certainty [15]. Such wide confidence intervals are not uncommon in sparse networks [3].

4.3. If the CI does not cross a threshold, we need to look at the effect size

Evidence from meta-epidemiologic studies [16-19], as well as numerous dramatic individual examples [20-25], have documented that large relative estimates of effects from relatively small randomized trials with few events are often refuted by subsequent trials. A systematic review and NMA of such early trials may result in a large pooled treatment effect and a CI that does not cross the thresholds, but which has wide CIs. The result will be dissemination of large overestimates of effect, and patients being exposed to unnecessary harms on the basis of overly optimistic estimates of treatment benefit.

When the relative effect size is modest (e.g,. less than 30% relative risk reduction or increase), a CI calculated based on a small number of patients is likely to be wide and cross a threshold, resulting in reviewers rating down for imprecision (Fig. 2, scenario 1a). When, however, the CI does not cross any threshold, reviewers should look at the size of the effect represented by the point estimate (Fig. 2, scenario 1b). We describe below two general cases to illustrate this concept. Because these are considerations relevant to the number of patients contributing information to a network estimate, typically calculated using relative estimates of effect, all assessments, and steps below focus on the relative estimate of effect (as opposed to the

assessments related to the thresholds, which are based on absolute estimates of effect).

4.3.1. When the effect size is modest and plausible, we should not rate down for imprecision

When the effect size is modest and plausible, a CI that is narrow and does not cross a relevant threshold, indicates a sufficient sample size underlying the network estimate (we will refer to this henceforth as the "effective sample size"). Therefore, when the CI does not cross a threshold and the effect size is modest, reviewers should not rate down due to imprecision (Fig. 1, scenario 2a).

For example, a systematic review comparing the effects of uterotonic agents for preventing postpartum hemorrhage [26] reported a risk ratio of 0.72 (95% CI, 0.56–0.97) for the comparison of carbetocin with oxytocin for the prevention of postpartum hemorrhage equal or larger than 500 milliliters. Because the CI does not cross the null effect and the size of the effect is modest, reviewers taking a minimally contextualized approach, and rating the certainty that there is a benefit of carbetocin over oxytocin would not rate down the certainty of this evidence due to imprecision. Indeed, 30,633 participants contributed to the direct estimate informing that comparison.

4.3.2. When the effect size is large, we should assess if the optimal information size is met

A CI may be wide and not cross a threshold only because the point estimate is large and potentially implausible. In this case, it is important to assess the effective sample size of the network estimate and determine if the OIS is met (Fig. 1, scenario 2b).

For example, in an NMA of methotrexate monotherapy vs. methotrexate combinations for patients with rheumatoid arthritis, the odds ratio associated with the comparison between methotrexate plus tofacinib vs. methotrexate alone was 3.0 with a 95% CI from 1.1 to 9.4. Even though the CI does not cross the threshold (i.e., null effect), because the effect size is large, reviewers should determine if the OIS is met before deciding whether to rate down for imprecision.

4.4. The confidence interval of a network estimate can help determine if the optimal information size is met and whether to rate down for imprecision

4.4.1. We can use the confidence interval of a network estimate to calculate the number of patients underlying that estimate

When presented with an estimate and CI for a treatment comparison from an NMA, we can inquire regarding the size of a single study that would give an estimate with that CI. This question puts aside the structure of the network and the combinations of direct and indirect evidence that are used to generate the network estimate and focuses on the estimate and its CI. With a small number of assumptions, it is possible to work backwards from the estimate of the treatment effect and its CI to calculate the equivalent sample size for the single study that would have generated those values (which we will call the "the effective sample size"). The details of the calculations depend on the type of outcome and effect measure. Boxes one, two, and three show the details for the RR, OR, and mean difference as treatment effect measures.

BOX 1.

The standard error of the log(RR) in a two-arm trial with equal arm sizes can be estimated as

$$SE_{trial} = SE(\log[RR]) = \sqrt{\frac{1}{n}\left(\frac{1}{p_c} + \frac{1}{RR \times p_c} - 2\right)}$$

Where n is the per-arm sample size, p_c is the observed proportion with a response in the reference group and RR is the observed relative risk with treatment.

The SE for the NMA estimate of the log(RR) can be calculated from the upper and lower limits of the reported CI

$$SE_{NMA} = \frac{\log(CI_{upper}) - \log(CI_{lower})}{3.92}$$

Setting these two estimates of the SE to be equal, we can solve for n as

$$n = \frac{\left(\frac{1}{p_c} + \frac{1}{RR \times p_c} - 2\right)}{(SE_{NMA})^2}$$

The value of p_c should be estimated from the relevant arms in the network, ideally with a meta-analysis of proportions. But as this is an approximate approach, it may be acceptable to simply count of the number of responses and number of patients across these arms to estimate p_c

BOX 2

The standard error of the log(OR) in a two-arm trial with equal arm sizes can be estimated as

$$SE_{trial} = SE(\log[OR]) = \sqrt{\frac{1}{n}\left(\frac{1}{p_c} + \frac{1}{1-p_c} + \frac{1}{p_t} + \frac{1}{1-p_t}\right)}$$

Where n is the per-arm sample size, p_c is the observed proportion with a response in the reference group and p_t is the observed proportion with a response in the reference group. Pt can be replaced by pt=pc*OR/(1-pc +pc*OR)

The SE for the NMA estimate of the log(OR) can be calculated from the upper and lower limits of the reported CI

$$SE_{NMA} = \frac{\log(CI_{upper}) - \log(CI_{lower})}{3.92}$$

Setting these two estimates of the SE to be equal, we can solve for n as

$$n = \frac{\left(\frac{1}{p_c} + \frac{1}{1-p_c} + \frac{1}{p_t} + \frac{1}{1-p_t}\right)}{(SE_{NMA})^2}$$

The value of p_c should be estimated from the relevant arms in the network, ideally with a meta-analysis of proportions. But as this is an approximate approach, it may be acceptable to simply count of the number of responses and number of patients across these arms to estimate p_c

BOX 3

The standard error of the mean difference in a two-arm trial with equal arm sizes and equal standard deviations in each group can be estimated as

$$SE_{trial} = SD \times \sqrt{\frac{2}{n}}$$

Where n is the per-arm sample size, and SD is the pooled within-group standard deviation

The SE for the NMA estimate of the mean difference can be calculated from the upper and lower limits of the reported CI

$$SE_{NMA} = \frac{CI_{upper} - CI_{lower}}{3.92}$$

Setting these two estimates of the SE to be equal, we can solve for n as

$$n = \frac{2 \times SD^2}{(SE_{NMA})^2}$$

The value of SD can be estimated as the pooled SD across all k arms relevant to the comparison

For example, an NMA of treatments to arrest dental caries [27] reported a network estimate of 0.67 for the RR comparing 1.23% APF gel to 5% NaF Varnish, with a 95% CI for the RR of 0.45–0.99. The overall risk of caries in the control groups (i.e., baseline risk) for this comparison was 77%. Using the formulas in Box one, we find that a single study with that observed RR, baseline risk and CI would have had 31 in each of the two arms, so the effective sample size of the network estimate is 62 patients. That same NMA reported a network estimate of 1.97 for the RR comparing 5% NaF varnish to usual care, with a 95%CI for the RR of 1.63–2.40. Using the formulas in Box one, we find that a single study with that observed RR, control group risk (37%) and CI would have 213 patients in each of the two arms, so the effective sample size of the network estimate is 426 patients.

4.4.2. If the effective sample size does not meet the optimal information size, we should rate down for imprecision

Focusing on the first example of the NMA of caries arrest (1.23% APF gel vs. 5% NaF Varnish), if reviewers judge the relative risk reduction of 33% to be large, they should assess whether the effective sample size of the data contributing to network estimate (62 patients) is at least as large as what we would require in a single trial: in other words, whether the OIS is met (Fig. 2, step 3).

A sample size calculation for a single trial, assuming that the reviewers consider a relative reduction of 25% (relative risk 0.75) as the minimal effect in arrest of caries to detect, with 80% power at alpha of 0.05, and a control group risk of 77% results in 92 patients per group or 184 in total. Therefore, the network estimate, with an effective sample size of 62, does not satisfy the OIS criterion. In a case like this, reviewers should rate down for imprecision (Fig. 2, scenario 3a)

The same process applies to the second example (5% NaF varnish vs. usual care). The overall risk of caries in the reference groups for this comparison was 37%. The sample size for a single trial with 80% power at alpha of 0.05 in which researchers want to detect a relative risk *increase* of 1 of 0.75 (so that the RR is 1.33) in the arrest of caries with a control group risk of 37% is 255 per group or 510 in total, so the network estimate with an effective sample size of 426, does not satisfy the OIS criterion and reviewers should rate down for imprecision (Fig. 2, scenario 3a). However, if reviewers determine that a relative increase of 40% is the minimum important effect, then the required single trial sample size would be 176 per group, the OIS criterion would be met, and reviewers should not rate down for imprecision (Fig. 1, scenario 3b).

Because the optimal information size depends on the minimal effect to detect used in the calculations, the same estimate may or may not meet the optimal information size based on the degree of contextualization and what this minimal effect is [28] Therefore, systematic reviewers who choose a biological plausible effect or a minimal important difference, and guideline developers who use a specific effect size based on full contextualization may come up with different judgments with regards to imprecision associated with optimal information size.

4.4.3. There may be scenarios in which calculating the effective sample size is not necessary

There may be scenarios in which the width of the CI is such that it is not necessary to calculate the effective sample size because the OIS will not be met. For relative risks and odds ratios, the CI width is represented by the ratio of the upper limit of the CI to the lower limit. When the ratio of the upper to the lower limits of the CI of the relative risk estimate is higher than three, the OIS of a RR estimate will not be met regardless of the effect size, any reasonable minimally important difference to detect, and baseline risk (appendix). In these cases, reviewers can confidently rate down due to imprecision without making any calculations.

In the two examples from the NMA of caries, these ratios are 2.2 (the result of dividing the upper limit of the CI, 0.99, by its lower limit, 0.45) and 1.5 (2.40 divided by 1.63), so the OIS calculation would be necessary. But in the example of methotrexate monotherapy vs. methotrexate combinations for patients with rheumatoid arthritis, the relative effect estimate was an odds ratio of 3.4, the upper limit of the CI was 9.4 and the lower limit was 1.1. The ratio of the upper to the lower limit of the CI is 9.4 of

1.1 = 8.5, so reviewers should rate down due to imprecision without the need to calculate the OIS.

When the OR is the effect measure, the guidelines are similar: if the ratio of the upper CI limit to the lower CI limit is larger than 2.5, in most cases the OIS criterion will not be met. If the ratio is in the neighborhood of three (e.g., 2.7–3.3) for the RR or 2.5 for the OR for a particular comparison, some reviewers may prefer to use the formulas in boxes one or two rather than this rule as this will clarify the degree to which OIS is met or not met.

For continuous outcomes, unfortunately, whether a CI is too wide to not have to calculate the effective sample size will vary importantly depending on the scale of the outcome, the use of standardized mean difference or mean difference as the effect measure, and the variability observed across studies. Thus, it is more straightforward for reviewers to calculate the effective sample size using the formulas in Box three.

4.5. When the NMA estimate does not meet the optimal information size, we can rate down the certainty of the evidence one or 2 levels

The width of the CI can guide the decision whether to rate down one or two levels due to imprecision. If reviewers had to calculate the effective sample size and determine if the OIS was met, this means that the CI is narrow enough to rate down one level only. In both examples of the treatments to arrest caries, if rating down, reviewers should rate down only one level. However, if the CI is wide enough to not have to bother calculating the effective sample size, reviewers may rate down two levels, as illustrated in the example of methotrexate monotherapy vs. methotrexate combination.

If the CI is extremely wide, reviewers should consider rating down 3 levels [15]. For example, in the first iteration of the living NMA of pharmacological treatments for Coronavirus Disease 2019 [14] for the comparison between Baloxavir Marboxil and standard care, the CI around the relative estimate of effect on mechanical ventilation ranged from 1.71 to 1.79×10^{26}. Such a CI does not require determining if the OIS is met: it is so wide that the uncertainty on the true estimate due to imprecision warrants rating down three levels.

5. Other considerations

Finally, previous GRADE guidance suggests that reviewers should avoid rating down the certainty of a network estimate when imprecision is associated with another limitation for which they already rated down: i.e., they should not double count. For instance, when imprecision is the result of serious incoherence between direct and indirect evidence, reviewers should only rate down for one of those domains [4].

6. Conclusion

Addressing imprecision of network estimates requires considerations of the level of contextualization, the target of the certainty rating, thresholds, how the CI relates to those thresholds, the magnitude of effect, and potentially the OIS. We have provided a previously unavailable approach to calculating the OIS in NMAs that systematic reviewers will certainly find useful.

7. Note

We provide an excel document with the supplementary material with worksheets allowing reviewers to easily calculate the OIS and effective sample size of a network estimate for binary outcomes using the estimate, the confidence interval, the risk in the control group, and an important difference between the groups. When the estimate of interest is the mean difference, the required inputs are the important difference between groups, the confidence interval and the pooled within-group standard deviation (which one can also calculated on our Excel worksheet by inputting data from relevant study arms).

8. Author contributions

RBP, GG, and GT conceptualized this work and collected examples. GT developed the statistical concepts related to the calculation of optimal information size in network estimates. RBP, GG, and GT drafted the manuscript. RM, DC, MH, and HS provided input that resulted in important modifications. All authors approved the final manuscript.

Acknowledgments

We would like to thank the authors of some of the systematic reviews we used as examples throughout this paper: Olivia Urquhart, and Glen Hazlewood. We would also like to thank all GRADE working group members for their input and feedback, in particular M Hassan Murad.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jclinepi.2021.07.011.

References

[1] Puhan MA, Schunemann HJ, Murad MH. A GRADE Working Group approach for rating the quality of treatment effect estimates from network meta-analysis. BMJ 2014;349:g5630. doi:10.1136/bmj.g5630.

[2] Brignardello-Petersen R, Bonner A, Alexander PE. Advances in the GRADE approach to rate the certainty in estimates from a network meta-analysis. J Clin Epidemiol 2018;93:36–44. doi:10.1016/j.jclinepi.2017.10.005.

[3] Brignardello-Petersen R, Murad MH, Walter SD. GRADE approach to rate the certainty from a network meta-analysis: avoiding spurious judgments of imprecision in sparse networks. J Clin Epidemiol 2019;105:60–7. doi:10.1016/j.jclinepi.2018.08.022.

[4] Brignardello-Petersen R, Mustafa RA, Siemieniuk RAC. GRADE approach to rate the certainty from a network meta-analysis: addressing incoherence. J Clin Epidemiol 2019;105:77–85. doi:10.1016/j.jclinepi.2018.11.025.

[5] Brignardello-Petersen R., Florez I., Izcovich A., GRADE approach to drawing conclusions from a network meta-analysis using a minimally contextualized framework. MBJ 2020;11(371):m3900 doi:10.1136/bmj.m3900

[6] Brignardello-Petersen R., Izcovich A., Rochwerg B., GRADE approach to making conclusions from a network meta-analysis using a partially contextualized framework. BMJ. 2020;10(371):m3907 doi:10.1136/bmj.m3907

[7] Guyatt GH, Oxman AD, Kunz R. GRADE guidelines 6. Rating the quality of evidence–imprecision. J Clin Epidemiol 2011;64(12):1283–93. doi:10.1016/j.jclinepi.2011.01.012.

[8] Guyatt G, Montori V, Devereaux PJ. Patients at the center: in our practice, and in our use of language. ACP J Club 2004;140(1):A11–12.

[9] Hultcrantz M, Rind D, Akl EA. The GRADE Working Group clarifies the construct of certainty of evidence. J Clin Epidemiol 2017;87:4–13. doi:10.1016/j.jclinepi.2017.05.006.

[10] Zeng L, Brignardello-Petersen R, Hultcrantz M. GRADE guidelines 32: GRADE offers guidance on choosing targets of GRADE certainty of evidence ratings. J Clin Epidemiol 2021;137:163–75. doi:10.1016/j.jclinepi.2021.03.026.

[11] Hazlewood GS, Barnabe C, Tomlinson G. Methotrexate monotherapy and methotrexate combination therapy with traditional and biologic disease modifying antirheumatic drugs for rheumatoid arthritis: abridged Cochrane systematic review and network meta-analysis. BMJ 2016;353:i1777.

[12] Florez ID, Veroniki AA, Al Khalifah R. Comparative effectiveness and safety of interventions for acute diarrhea and gastroenteritis in children: A systematic review and network meta-analysis. PLoS One 2018;13(12):e0207701. doi:10.1371/journal.pone.0207701.

[13] Santesso N, Glenton C, Dahm P. GRADE guidelines 26: Informative statements to communicate the findings of systematic reviews of interventions. J Clin Epidemiol 2020;119:126–35. doi:10.1016/j.jclinepi.2019.10.014.

[14] Siemieniuk R, Bartoszko JJ, Ge L. Drug treatments for covid-19: living systematic review and network meta-analysis. Bmj 2020;370:m2980.

[15] Piggott T, Morgan RL, Cuello-Garcia CA. Grading of recommendations assessment, development, and evaluations (GRADE) notes: extremely serious, GRADE's terminology for rating down by three levels. J Clin Epidemiol 2020;120:116–20. doi:10.1016/j.jclinepi.2019.11.019.

[16] Pereira TV, Horwitz RI, Ioannidis JP. Empirical evaluation of very large treatment effects of medical interventions. JAMA 2012;308(16):1676–84. doi:10.1001/jama.2012.13444.

[17] Rerkasem K, Rothwell PM. Meta-analysis of small randomized controlled trials in surgery may be unreliable. Br J Surg 2010;97(4):466–9. doi:10.1002/bjs.6988.

[18] Trikalinos TA, Churchill R, Ferri M. Effect sizes in cumulative meta-analyses of mental health randomized trials evolved over time. J Clin Epidemiol 2004;57(11):1124–30. doi:10.1016/j.jclinepi.2004.02.018.

[19] Nagendran M, Pereira TV, Kiew G. Very large treatment effects in randomised trials as an empirical marker to indicate whether subsequent trials are necessary: meta-epidemiological assessment. BMJ 2016;355:i5432.

[20] Desai K, Carroll I, Asch S. Extremely large outlier treatment effects may be a footprint of bias in trials from less developed countries: randomized trials of gabapentinoids. J Clin Epidemiol 2019;106:80–7. doi:10.1016/j.jclinepi.2018.10.012.

[21] Bosch J, Yusuf S, Gerstein HC. Effect of ramipril on the incidence of diabetes. N Engl J Med 2006;355(15):1551–62. doi:10.1056/NEJMoa065061.

[22] Devereaux PJ, Beattie WS, Choi PT. How strong is the evidence for the use of perioperative beta blockers in non-cardiac surgery? Systematic review and meta-analysis of randomised controlled trials. BMJ 2005;331(7512):313–21.

[23] Bangalore S, Wetterslev J, Pranesh S. Perioperative beta blockers in patients having non-cardiac surgery: a meta-analysis. Lancet 2008;372(9654):1962–76. doi:10.1016/s0140-6736(08)61560-3.

[24] Yusuf S, Collins R, MacMahon S. Effect of intravenous nitrates on mortality in acute myocardial infarction: an overview of the randomised trials. Lancet 1988;1(8594):1088–92. doi:10.1016/s0140-6736(88)91906-x.

[25] Finfer S, Bellomo R, Boyce N. A comparison of albumin and saline for fluid resuscitation in the intensive care unit. N Engl J Med 2004;350(22):2247–56. doi:10.1056/NEJMoa040232.

[26] Gallos ID, Papadopoulou A, Man R. Uterotonic agents for preventing postpartum haemorrhage: a network meta-analysis. Cochrane Database Syst Rev 2018;12(12):CD011689.

[27] Urquhart OA-O, Tampi MP, Pilcher L. Nonrestorative treatments for caries: systematic review and network meta-analysis. J Dent Res 2019;98(1):14–26.

[28] Schunemann HJ. Interpreting GRADE's levels of certainty or quality of the evidence: GRADE for statisticians, considering review information size or less emphasis on imprecision? J Clin Epidemiol 2016;75:6–15. doi:10.1016/j.jclinepi.2016.03.018.